

LABORATORY MEDICINE PROGRAM

UUID: 5595A0A9-FD0D-4F79-8DE5-28DC37C96C68

TCGA-EM-A303-01A-PR

Redacted



Surgical Pathology Consultation Report

Accession

Collector

Receiver

Referral

Criteria

Diagnosis Discrepancy

Yes

No

Primary Tumor Site Discrepancy

HiPAA Discrepancy

Prior Malignancy History

Dual/Synchronous Primary Noted

Case # (single)

QUALIFIED

DISQUALIFIED

Reviewer Initials

Date reviewed: 12/11/12

Specimen(s) Received

1. Parathyroid: Query Rt Inferior Parathyroid
2. Thyroid: Total thyroid gland-long stitch marks Lt. superior pole, short stitch marks Rt. superior pole
3. Lymph node: Rt. Paratracheal nodes
4. Lymph node: Lt. paratracheal nodes and upper mediastinal

ICD-O-3
carcinoma, papillary, thyroid 8260/3
Site: thyroid, NOS C73.9
12/15/12

Diagnosis

1. No pathological diagnosis: Lymph node (listed as 'query right inferior parathyroid') excisional biopsy
2. Widely-invasive papillary carcinoma with unusual features, 3.1 cm, right lobe and isthmus; underlying severe chronic lymphocytic thyroiditis with fibrosis and atrophy: Thyroid
No pathological diagnosis: Lymph nodes, 4, isthmic
- Total thyroidectomy specimen. See Comment.
3. Metastatic papillary thyroid carcinoma: Lymph nodes, 2 of 8 (right paratracheal) dissection specimen
4. Metastatic papillary thyroid carcinoma: Lymph node, 1 of 3 (left paratracheal and upper mediastinal) dissection specimen

Comment

The specimen reveals a widely-invasive classical variant papillary thyroid carcinoma that exhibits areas of Warthin-like and hobnail cell change. There is also focal solid growth with cribriform and morular/squamoid architecture, which accounts for less than 20% of the lesion and is characterized by increased mitotic activity (5/10 HPF) and apoptotic change. These areas are positive for TTF-1 and thyroglobulin. Although some of these features may represent microscopic dedifferentiation within this lesion, these features are unusual and therefore the clinical significance of these findings is not well known.

One of the focal components observed in this lesion is the cribriform and morular architecture, which is characteristic of lesions that have been reported in association with the Familial Adenomatous Polyposis (FAP) syndrome. Although this is not diagnostic of this entity, further clinical evaluation is recommended.

Dr. [redacted] has also reviewed this case and she agrees with the diagnosis and interpretation.

Synoptic Data

Procedure:

Other: Total thyroidectomy with right paratracheal and left paratracheal & upper mediastinal dissection

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Received: Fresh
Specimen Integrity: Intact
Specimen Size: Right lobe:
3.4 cm
1.6 cm
0.8 cm
Left lobe:
3.4 cm
1.7 cm
0.6 cm
Isthmus +/- pyramidal lobe:
8.5 cm
3.0 cm
1.6 cm

Specimen Weight: 14.5 g
Tumor Focality: Unifocal

----- DOMINANT TUMOR -----

Tumor Laterality: Right lobe
Isthmus

Tumor Size: Greatest dimension: 3.1cm
Additional dimension: 2.7 cm
Additional dimension: 1.4 cm

Histologic Type: Papillary carcinoma, classical (usual)
Papillary carcinoma, other variant: with focal Warthin-like and hobnail cell change, and
cribriform-morular architecture
Classical (papillary) architecture
Classical cytomorphology

Histologic Grade: Not applicable
Margins: Margins uninvolved by carcinoma

Distance of invasive carcinoma to closest margin: 0.1mm
Partially encapsulated

Tumor Capsule: Partially encapsulated

Tumor Capsular Invasion: Present, extent widely invasive

Lymph-Vascular Invasion: Not identified

Perineural Invasion: Not identified

Extrathyroidal Extension: Not identified

TNM Descriptors: Not applicable

Primary Tumor (pT): pT2: Tumor more than 2 cm, but not more than 4 cm, limited to thyroid

Regional Lymph Nodes (pN): pN1a: Nodal metastases to Level VI (pretracheal, paratracheal and
prelaryngeal/Delphian) lymph nodes

Number of regional lymph nodes examined: 16

Number of regional lymph nodes involved: 3

Lymph Node, Extranodal Extension Not identified

Distant Metastasis (pM): Not applicable

Additional Pathologic Findings: Thyroiditis, advanced

Other: marked thyroid atrophy

Ancillary Studies: Type: Immunohistochemistry

Results: The tumor is positive for thyroglobulin, TTF-1, HBME-1, and is negative for
calcitonin, CD31 and CD5

*Pathologic Staging is based on AJCC/UICC TNM, 7th Edition

Electronically verified by:

Surgical Pathology Consultation Report

Gross Description

1. The specimen labeled with the patient's name and as "query right inferior parathyroid", contains one tiny piece of pink tissue that measures 0.2 x 0.1 x 0.1 cm. The specimen is submitted for intraoperative consultation in toto.
1A frozen tissue resubmitted

2. The specimen labeled with the patient's name and as "total thyroid gland, long stitch marks left superior pole, short stitch marks right superior pole" consists of a thyroid gland that is oriented with a long suture at the left superior pole and a short suture at the right superior pole and weighs 14.5 g. The right lobe measures 3.4 cm SI x 1.6 cm ML x 0.8 cm AP, the left lobe measures 3.4 cm SI x 1.7 cm ML x 0.6 cm AP and the isthmus measures 8.5 cm SI x 3.0 cm ML x 1.6 cm AP. The external surfaces have fibrous adhesions and are painted with silver nitrate. No parathyroid glands or no lymph nodes are identified grossly. The isthmus and lower pole of the right lobe contains a pale tan well delineated nodule that measures 3.1 cm SI x 2.7 cm ML x 1.4 cm AP. The remainder of the thyroid tissue is unremarkable. Two sections of the nodule and one section of normal thyroid tissue are stored frozen. The remainder of the specimen is submitted as follows:

2A-E right lobe submitted in toto, superior to inferior

2F-M isthmus submitted in toto from right to left

2N-P left lobe, submitted in toto, superior to inferior

3. The specimen labeled with the patient's name and as "right paratracheal nodes", contains 6 tan fatty lymph nodes that measure from 0.4 x 0.2 x 0.3 up to 1.9 x 0.8 x 0.4 cm.

3A-B a single lymph node bisected in each block

3C 2 lymph nodes bisected

3D a single lymph node bisected

3E 2 individual lymph nodes

4. The specimen labeled with the patient's name and as "left paratracheal node and upper mediastinal", contains 4 tan fatty lymph nodes that measure from 0.6 x 0.4 x 0.4 cm up to 0.9 x 0.4 x 0.4 cm.

4A lymph nodes submitted in toto

Quick Section Diagnosis

1. Query parathyroid: Lymphoid tissue. No parathyroid identified. . reported at

Microscopic Description

MD

The specimen reveals a widely-invasive classical variant papillary carcinoma that contains areas of Warthin-like and hobnail cell change. There is also focal solid growth with cribriform and morular-like squamoid architecture. These areas exhibit apoptotic changes and increased mitotic activity (5/10 HPF, especially on slide 2D). The tumor is positive for TTF-1, thyroglobulin, HBME-1 (apical and membranous), vimentin and monoclonal CEA (focal), and is negative for CD5, calcitonin and CD31. p53 stains less than 3% of tumor cell nuclei.